

From Space Technology to Cosmology: A 30-Year Analysis of NASA’s Astronomy Picture Of the Day

Using APOD Posts to Study Changes in NASA’s Public
Astronomy Communication

Thiago Schuck

May 2026

Abstract

In 1995, NASA began posting daily pictures related to astronomy and physics, along with a short explanation on a new website called Astronomy Picture of the Day (APOD). Since then, several discoveries in astronomy and physics have been made. This report scrapes and analyzes APOD posts from the first post in 1995 to the present to examine how the topics have changed over time. Using the title and explanation text from each APOD post, TF-IDF features and a Naive Bayes classifier were applied to assign topic labels from the APOD’s index page to each entry. Then, after cleaning, analyzing and visualizing the labeled dataset, the results show that APOD’s yearly topic distribution changed from a balanced mix of Cosmology, Space Technology, and Solar System, towards Cosmology becoming the most common category. However, Cosmology’s distribution remained comparable to the combined distribution of Space Technology and Solar System, while Sky-related posts account for the majority of the remaining posts. Overall, APOD’s focus shifted away from Space Technology to more Cosmology, Solar System, and Sky-related posts, suggesting an increase in emphasis on natural astronomical objects and phenomena rather than human-made technology.

1 Introduction

NASA’s Astronomy Picture Of the Day is a website where NASA has posted images related to astronomy and physics with a short explanation since 1995. A typical post features an image from a space telescope, like Hubble or the James Webb Space Telescope; or from an observatory, such Mount Wilson Observatory; or even a picture of a famous scientist like Isaac Newton.

APOD can offer insight into NASA’s public-facing communication priorities while serving as a tool for public education and inspiration. With over 10,000 entries covering a variety of different astronomical phenomena and topics, each

with its own explanations, APOD is a valuable source of learning and research. Because APOD has been updated daily for almost 30 years, it is a useful dataset for analyzing how NASA’s public communication has changed over time.

This project asks how the topic distribution of APOD posts has changed since the first post in 1995. To answer this question, I scraped each APOD post, extracting the title, explanation text, and image using NASA’s open API. I then used TF-IDF features and a Naive Bayes classifier to assign topic labels to each entry, along with a confidence score for each label. After removing the low confidence entries, I visualized the data with several different graph types. The results suggest that APOD’s focus has shifted away from Space Technology and towards an emphasis on Cosmology, the Solar System, and Sky-related posts.

2 Related Work

2.1 Astronomy Picture Of the Day and Science Communication

NASA’s APOD is a long-running archive that pairs images of astronomy-related images with short explanations for a general audience. Unlike raw data, APOD showcases astronomy through selected images and easily understandable descriptions, making it useful to study NASA’s public-facing science communication. However, it is not an accurate representation of NASA’s current goals and research priorities.

2.2 Text Classification Methods

2.2.1 TF-IDF

Because the APOD posts are written as natural language text, the title and explanation of each post had to be converted into numerical features before classification. To do this, this project uses term frequency-inverse document frequency (TF-IDF). TF-IDF assigns higher weights to words or phrases that are important within a specific document but less common across the full dataset. This is useful for APOD because common words such as “the” or “image” are less helpful for classification, while terms such as “galaxy,” “nebula,” “spacecraft,” or “eclipse” are more useful for distinguishing topics.

In this project, the title and explanation text were combined into one document for each APOD entry. The TF-IDF vectorizer removed English stop words, included both single words and two-word phrases, and ignored terms that appeared in fewer than two documents. This created a feature matrix that represented each APOD post based on important and distinctive words and phrases.

2.2.2 Naive Bayes Classification

After converting the APOD text into TF-IDF features, this project uses a Multinomial Naive Bayes classifier to assign topic labels to each post. Naive Bayes is commonly used for document classification because it estimates the probability that a document belongs to a category based on the words it contains. Although the model assumes that features are independent, which is not completely true for natural language, it is still effective for many text classification tasks.

2.3 Visualizing Topic Trends Over Time

After text classification, visualization is needed to interpret how APOD topic labels change over time. Since each APOD entry has both a date and a topic label, the data can be grouped by year or multi-year periods to show changes in category frequency. This makes it possible to compare dominant topics across different time periods and identify long-term shifts in APOD’s public-facing emphasis.

This project uses visualizations such as heatmaps, pie charts, and other time-based distribution plots to show these changes. Heatmaps are useful for comparing how the frequency of multiple categories changes over time, while pie charts and percentage-based plots show how much of the yearly distribution each category represents. Together, these visualizations help show both the overall distribution of APOD topics and the changes in frequency for each category over time.

3 Methodology

3.1 Data Collection

Using NASA’s open API, I collected APOD entries from the official APOD API endpoint, beginning with the first APOD post on June 16, 1995. For each entry, I collected the available metadata fields, including the title, date, explanation, media type, image URL, high-definition image URL, thumbnail URL, copyright information, and service version. Using Python and pandas, I saved the collected data as a CSV file in the project’s data folder.

3.2 Data Cleaning and Processing

3.2.1 Topic Labeling

The classifier was trained using weak labels generated from keyword matches based on APOD’s index categories. These weak labels provided an initial set of labeled examples, and the Naive Bayes model was then used to generalize beyond exact keyword matches. For each APOD entry, the model predicted a topic label and a confidence score.

3.2.2 Confidence Filtering

After each APOD entry received a predicted label and confidence score, entries with low confidence were removed before visualization. This reduced the number of uncertain classifications included in the final analysis and allowed the visualizations to focus on more reliable topic labels.

3.2.3 Missing Value Handling

Before creating the visualizations, I removed missing values that could interfere with plotting. Since the visualizations compare APOD topic categories over time, the most important fields were the date column and the final category label column. After confidence filtering, the visualization functions checked that these required columns existed, converted the date column into a datetime format, and removed entries with missing dates or missing category labels.

Missing values in other columns, such as image URLs or copyright information, were not removed for the visualization stage because those fields were not used in the topic trend plots. This allowed the dataset to keep entries with incomplete metadata while still ensuring that every plotted entry had both a valid date and a valid topic label.

3.3 Data Visualization

Using Python, matplotlib, pandas, and math, I created several visualizations to examine how APOD topic categories changed over time. These visualizations were designed to show both the raw frequency of each category and the percentage distribution of categories across years and larger time periods.

3.3.1 Frequency Heatmap

I created a heatmap to show the frequency of each category across years. Each category-year cell includes a numerical count, making it easier to identify peaks, dips, and long-term changes in category frequency.

3.3.2 Yearly Pie Charts

I created annual pie charts showing the percentage distribution of APOD categories for each year. These charts make it possible to compare the dominance of each topic in a given year.

3.3.3 Distribution Line Graph

I combined the Cosmos subcategories into one parent Cosmology category and created a line graph showing each category's yearly percentage distribution. The original year-by-year graph was noisy, making long-term trends difficult to identify. To reduce this noise, I also created a three-year rolling average version of the graph. This made the overall trends easier to read while still preserving major peaks and dips.

3.3.4 Five-Period Distribution Table

Using the combined Cosmology category, I divided the dataset into five six-year periods and calculated the average percentage distribution of each category within each period. This table makes it easier to compare broad changes in APOD’s topic distribution over time and shows how each category’s share changed across the full archive.

4 Experimentation & Results

4.1 Category Frequency Over Time

Figure 1 shows the frequency of each APOD category over time. This graph was used to identify which categories appeared most often and whether certain categories increased or decreased across the APOD archive.

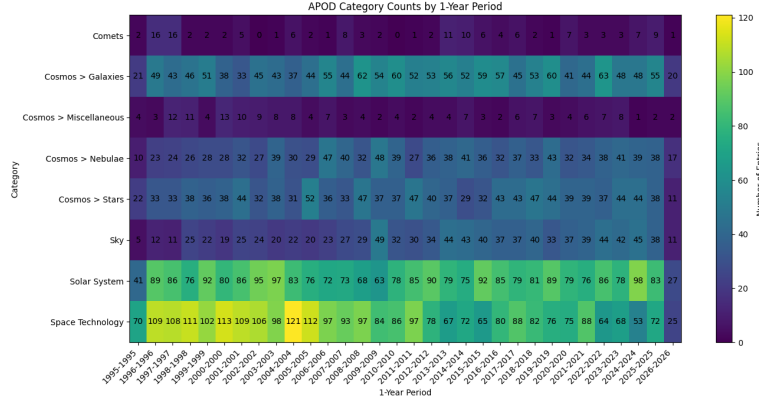


Figure 1: APOD category frequency heatmap over time.

Space Technology and Solar System are the most prevalent categories; if you combine all Cosmos categories, it will also become one of the dominant categories. Sky starts rare and steadily climbs to almost 400% in 2025. Every category in both 1995 and 2026 is shown as darker because those are incomplete years.

4.2 Yearly Category Distribution

Figure 2 and Figure 3 show the yearly distribution of APOD categories. These graphs were used to show APOD’s topic focus each year and how it changed over time.

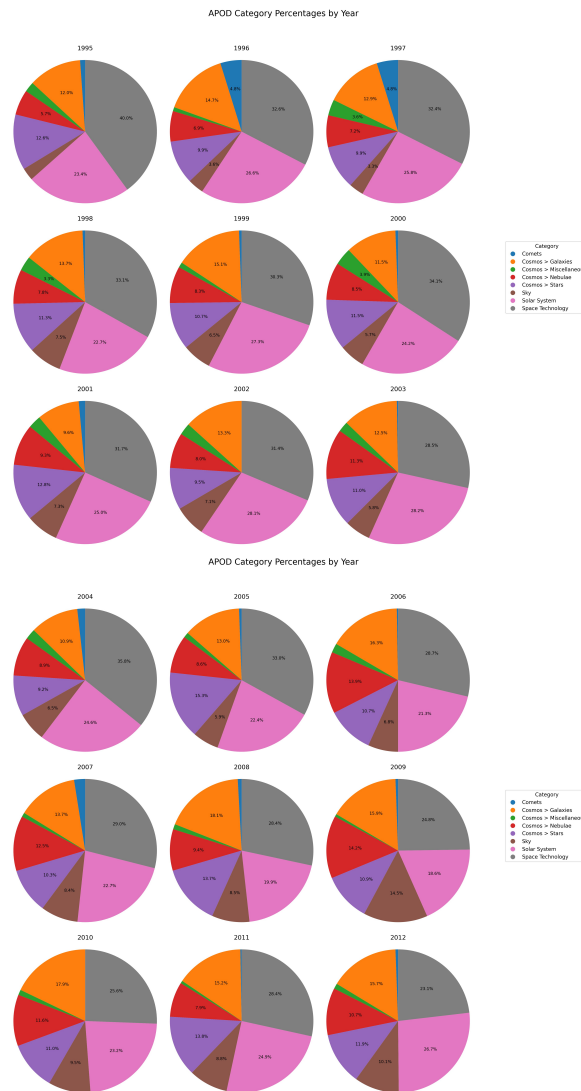


Figure 2: Yearly APOD category distribution pie charts, first half of the dataset.

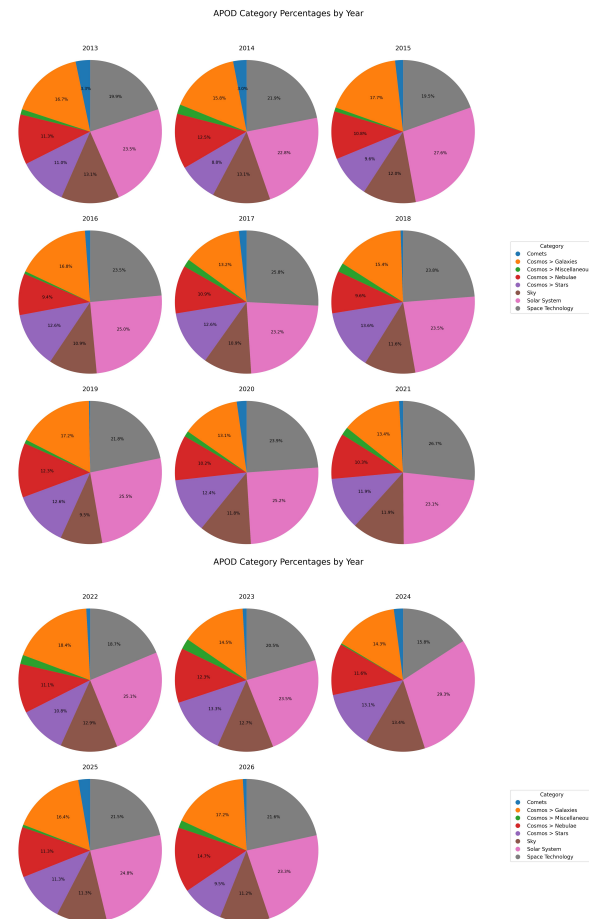


Figure 3: Yearly APOD category distribution pie charts, second half of the dataset.

The pie charts in the early APOD years show the distribution favored Space Technology and Solar System (roughly 2/3), with the majority of the remaining distribution being the Cosmos sub-categories. In 2006, the distribution shifted towards Space Technology and Solar System, being around 50% of the distribution, and the Cosmos sub-categories slightly increasing. Then, in 2013, the distribution shifted again towards Space Technology and Solar System to be roughly equal to the Cosmos sub-categories at around 40%. Sky has also been increasing since around 2006, and now covers the majority of what the dominant categories do not.

4.3 Distribution Trends Over Time

Figure 4 shows the distribution of each category over time, with Cosmology being the parent category for all Cosmos subcategories. This graph was used to show which categories increased and decreased relative to each other.

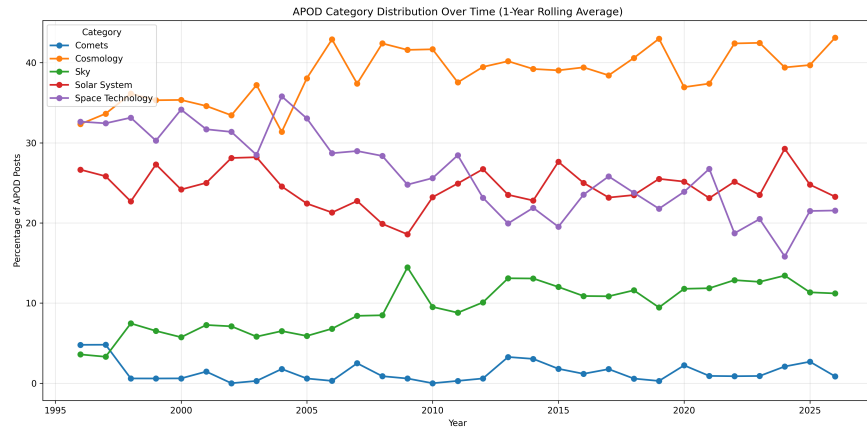


Figure 4: APOD category distribution over time without smoothing.

The graph shows many fluctuations, but it is still clear that before 2005, the Cosmology, Solar System, and Space Technology categories were close in frequency. After 2005, Space Technology decreases, while Cosmology increases and Solar System stays relatively stable.

Figure 5 shows the same distribution using a three-year rolling average to smooth out the lines and improve readability.

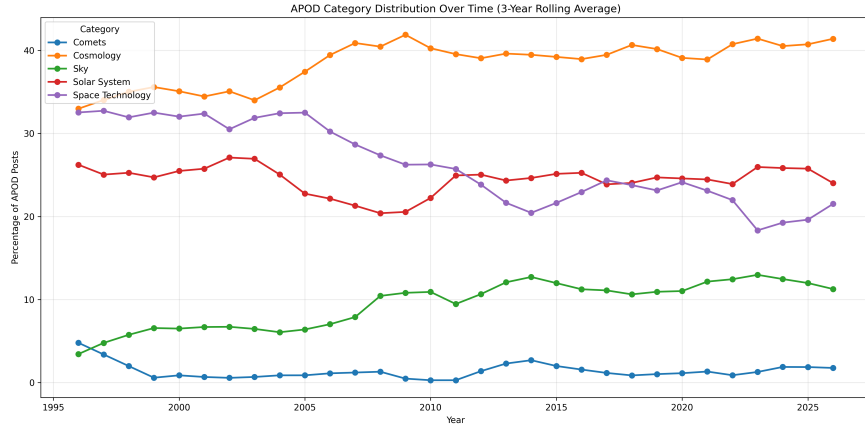


Figure 5: APOD category distribution over time using a three-year rolling average.

Using a three-year rolling average, the trends of each category become clearer. Cosmology rose to around 40% in 2007 and has stayed around that level through the present. Space Technology continuously decreases, and after 2007, rather than Cosmology taking the would-be Space Technology posts, Sky becomes more frequent. The Solar System category has large fluctuations, but does not show a consistent pattern of decreasing or increasing.

4.4 Period-Based Topic Comparison

Figure 6 splits the APOD dataset into 5 and averages the distributions for each category. This graph was used to find how much and where each category distribution changed.

APOD Category Percentage Distribution Across 5 Time Periods

	1996-2002	2003-2009	2010-2016	2017-2023	2024-2026
Cosmology	34.4%	38.7%	39.5%	40.2%	40.1%
Space Technology	32.2%	29.7%	23.2%	23.0%	19.1%
Solar System	25.7%	22.5%	24.8%	24.1%	26.5%
Sky	5.9%	8.0%	11.1%	11.6%	12.2%
Comets	1.8%	1.0%	1.4%	1.1%	2.2%

Figure 6: APOD average distribution split into 5 periods

This table chart offers an overview of exactly how each category has changed

through larger time periods. Cosmology shows a large increase of 4.3% from group 1996-2002 to 2003-2009, with a total increase of 5.7% over all 30 years. Space Technology has its largest decrease of 6.5% in 2003-2009 to 2010-2016 and has by far the largest overall decrease of 13.1%. Solar System fluctuates by 4% over 30 years but starts and ends at a similar frequency. Sky has its largest increase of 3.1% the same years as Space Technology's largest decrease from 2003-2009 to 2010-2016. Finally, Comets stays roughly the same throughout with it only fluctuating 1.2%.

4.5 Summary of Main Findings

Based on APOD's index categories and sample posts from each category, the Sky category largely represents constellations, night-sky views, and other astronomical phenomena visible from Earth. Therefore, the APOD categories can be broadly grouped into two larger themes: astronomical phenomena and engineering. The astronomical phenomena group includes Cosmology, Solar System, Sky, and Comets, while the engineering group is represented by Space Technology.

This grouping shows that NASA's public communication through APOD has consistently focused mainly on astronomical phenomena rather than engineering. However, this focus has become even stronger over time. Space Technology decreased by about 10% , roughly one-fourth of its original share, of the yearly distribution over the full APOD archive, while the categories related to astronomical phenomena became more dominant. If the current pattern continues, Space Technology may continue to represent a smaller portion of APOD's yearly topic distribution.

5 Limitations

This report shows an interesting aspect of NASA's public-facing communication through APOD, but there are several important limitations involving both the methodology and APOD itself.

1. APOD has another category that is not included in this report: People. This category contains posts about astronauts, scientists, and other important figures in astronomy. During data cleaning and processing, I manually checked the lowest-confidence entries and found that the Naive Bayes classifier did not accurately classify people-related posts. Therefore, rather than keeping an inaccurate category that would add noise to the analysis, I removed it from the final results. However, because the classifier assigned many People posts low confidence scores, most of these entries were likely removed during confidence filtering. Furthermore, because this project uses TF-IDF features and a Naive Bayes classifier, it would be incorrect to claim that every topic label is 100% accurate.

2. Because all APOD posts are related to astronomy, many entries could reasonably belong to multiple categories. For example, a post about a telescope image of Jupiter could relate to Space Technology, the Solar System, and Sky depending on how the post is interpreted. Therefore, the category labels should be understood as best-fit labels rather than perfectly distinct topics.
3. APOD represents only one part of NASA. It does not include all mission timelines, future work, internal priorities, or current frontier research. Therefore, this report cannot make broad claims about NASA’s overall goals or research interests. Instead, the results should be interpreted as changes in how NASA is communicated through APOD only.

6 Conclusion

This report examined almost 30 years, and over 11,100 posts about astronomy on NASA’s Astronomy Picture Of the Day website and how it has changed from its first post in 1995. By scraping APOD entries, extracting each post’s title and explanation text, and applying TF-IDF features with a Naive Bayes classifier, this report found changes and patterns in APOD’s topic distribution.

The results show that APOD has consistently focused mainly on astronomical phenomena rather than human-made space technology. Categories such as Cosmology, Solar System, and Sky, make up most of the dataset across the full archive. However, the balance between categories changed over time. APOD had a heavy emphasis on Space Technology in 1995, but in 30 years it’s lost over 10%, or a quarter of it’s distribution. At the same time, Cosmology became the largest individual category, while Sky and Solar System posts also remained important parts of the distribution.

Overall, APOD’s primary focus has always been astronomy. However, over the last 30 years, there has been a redistribution within APOD’s existing focus, with less emphasis on Space Technology and more emphasis on natural astronomical phenomena. Because APOD is only one part of NASA’s public-facing communication, this trend cannot be interpreted as NASA as a whole moving away from engineering. Instead, it suggests that, through daily APOD posts, NASA has increasingly emphasized educating the public about natural objects and phenomena in the universe beyond human-made technology.

7 Use of AI

Throughout this project I worked with AI. I gave the AI prompts to create each feature in the code, and then I revised it, corrected any errors, or gave another prompt to further specify what I wanted in this project. I also used it to learn more about LaTeX code and to check for any grammar errors in my report.

Throughout this project, I used AI as a support tool for coding, debugging, LaTeX formatting, and report revision. I prompted the AI to help create fea-

tures in the code, then reviewed, revised, and corrected the output as needed. When the generated code did not fully match the goals of the project, I modified it myself or gave more specific prompts to improve the result.

I also used AI to learn more about LaTeX syntax and to check the grammar and clarity of the written report. The final code, analysis decisions, and visualizations were reviewed and edited by me. While AI was used as a support tool, the interpretation of results and final report content were written by me.

8 References

1. NASA. *Astronomy Picture of the Day Index - Main Page*. Available at: <https://apod.nasa.gov/apod/lib/aptree.html>
2. NASA. *Astronomy Picture of the Day Archive*. Available at: <https://apod.nasa.gov/apod/archivepixFull.html>
3. NASA. *Astronomy Picture of the Day API*. Available at: <https://api.nasa.gov/>
4. Schuck, Thiago. *APOD Topic Analysis Project Code*. GitHub repository. Available at: <https://github.com/ThiagoS04/30-year-apod-analysis>